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A Parameter Study of Transonic Airfoil Flutter and Limit Cycle Oscillation Behavior

Denis B. Kholodar, * Jeffrey P. Thomas, † Earl H. Dowell, ‡ and Kenneth C. Hall §
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Using a state of the art inviscid computational fluid dynamic (CFD) harmonic balance (HB) flow solver a parametric investigation of how structural parameters and freestream Mach number of a transonic flow affect the flutter and limit cycle oscillation (LCO) characteristics of a "typical" two degrees-of-freedom (DOF) transonic airfoil configuration is presented. The computational efficiency of the harmonic balance aerodynamic model allows a much more thorough exploration of the parameter range than has been possible previously.

Nomenclature

a = nondimensional location of airfoil elastic axis, $a = \frac{e}{b}$
b, c = semi-chord and chord, respectively
$\bar{c}_l, \bar{c}_m$ = nondimensional coefficients of lift and moment about elastic axis for simple harmonic motion
$\bar{c}_{l_{\bar{h}}}, \bar{c}_{l_{\bar{\alpha}}}$ = nondimensional coefficients of lift due to plunging and pitching motions in time-linearized aerodynamics
$\bar{c}_{m_{\bar{h}}}, \bar{c}_{m_{\bar{\alpha}}}$ = nondimensional coefficients of moment due to plunging and pitching motions in time-linearized aerodynamics
e = location of airfoil elastic axis, measured positive aft of airfoil midchord
h, α = plunge and pitch DOF, respectively
$\bar{h}, \bar{\alpha}$ = nondimesional amplitudes of plunge, $\bar{h} = \frac{h}{b}$, and pitch harmonic excitations, respectively
I_α = second moment of inertia about elastic axis
$L, M_{\text{e.a.}}$ = lift and moment about elastic axis
K_h, K_α = airfoil plunge stiffness and torsional stiffness about elastic axis
M = freestream Mach number
m = airfoil sectional mass
μ = mass ratio, $\mu \equiv \frac{m}{\pi \rho_\infty b^2}$
N = number of harmonics used in HB solver
N_r, N_Θ = number of grid points in radial and circumferential direction, respectively
R = radius of computational domain
r_α = radius of gyration of airfoil about elastic axis, $r_\alpha^2 \equiv \frac{I_\alpha}{mb^2}$
S_α = first moment of inertia about elastic axis
U_∞ = freestream velocity
V = reduced velocity, $V \equiv \frac{U_\infty}{\omega_\alpha b}$
$\frac{V_f}{\sqrt{\mu}}$ = flutter speed index, $\frac{V_f}{\sqrt{\mu}} \equiv \frac{U_{\infty f}}{\sqrt{\mu}\,\omega_\alpha b}$
$\omega, \bar{\omega}$ = frequency and reduced frequency based on airfoil chord, $\bar{\omega} \equiv \frac{\omega c}{U_\infty}$
ω_α, ω_h = uncoupled natural frequencies of pitch and plunge DOF
ω_1, ω_2 = coupled structural natural frequencies
x_α = airfoil static unbalance, $x_\alpha \equiv \frac{S_\alpha}{mb}$

Introduction

Transonic flow flutter and limit cycle oscillations (LCO) are of significant interest in wing and aircraft design. The large expense incurred in both time- and frequency-domain transonic aerodynamic computations is the principal obstacle to the aeroelastician in obtaining a deeper understanding of these phenomena through a systematic parameter study.

Reduced order modeling (ROM) techniques have been developed and used to overcome this obstacle in recent work on this subject. For a general review of the latest studies involving ROM based methods see Ref. [1]. Pade approximants of transfer functions and various other rational polynomial curve fitting techniques also have been used in recent years for linear flutter analysis, e.g. see Ref. [2–4].

In the past few years at Duke University a number of CFD time (dynamically) linearized codes have been developed[5,6] and converted to the frequency domain. ROM techniques have then been applied to these dynamically linearized CFD codes and then used for flutter analyses that are very computationally efficient.

Recently, a novel nonlinear harmonic balance (HB) method that extends the frequency domain CFD models to the fully dynamically nonlinear range has been developed. This method enables one to model efficiently nonlinear unsteady aerodynamic behavior corresponding to finite amplitude structural motion of a

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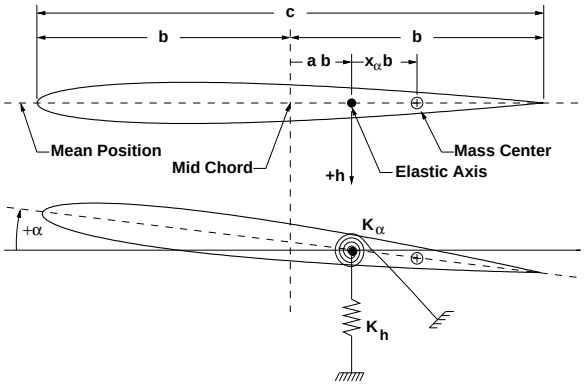


Fig. 1 Sketch of Aeroelastic Typical Section.

prescribed frequency, and which can be subsequently used for modeling LCO behavior.[5,6] We believe these two methods, ROM based flutter analysis and HB based LCO modeling, will significantly advance the aeroelastician's efforts to do rapid parametric studies.

As was first demonstrated in Ref. [6], after determining the flutter boundary of a "typical" airfoil aeroelastic configuration, a subsequent LCO study of the same model undergoing large motions can be conducted with aid of the HB method.[5,6] The flutter and LCO study described here is a continuation and extension of the work done in Ref. [6] with a more thorough investigation of the influence of structural parameters and transonic Mach number on LCO.

Governing Equations

Consider a "typical" two degrees-of-freedom (DOF) airfoil section as shown in Fig. 1. The equations of motion of this aeroelastic system can be written in the form:

$$\begin{aligned} m\ddot{h} + S_\alpha \ddot{\alpha} + K_h h &= -L \\ S_\alpha \ddot{h} + I_\alpha \ddot{\alpha} + K_\alpha \alpha &= M_{\text{e.a.}} \end{aligned} \tag{1}$$

Here the left-hand side terms represent a linear structural model approximation for the plunge and pitch coordinates. The right hand side terms represent the aerodynamic loading terms, which for this study are based upon the harmonic balance approach applied to a discrete CFD model of the inviscid Euler equations. Here we present a summary of the method. For a more detailed description see Ref. [5,6].

In integral form, the unsteady Euler equations may be written as

$$\begin{aligned} \frac{\partial}{\partial t}\iint_{D(t)} \mathbf{U}\,\mathrm{d}D + \oint_{\partial D(t)} \left(\mathbf{F}(\mathbf{U}) - \mathbf{U}\frac{\partial f}{\partial t}\right)\mathrm{d}y \\ - \oint_{\partial D(t)} \left(\mathbf{G}(\mathbf{U}) - \mathbf{U}\frac{\partial g}{\partial t}\right)\mathrm{d}x = \mathbf{0} \end{aligned} \tag{2}$$

where x and y are the Cartesian coordinates, t is time, and D is a deforming control volume bounded by the control surface ∂D. The quantities $\frac{\partial f}{\partial t}$ and $\frac{\partial g}{\partial t}$ are the x and y components of the velocity of the control surface ∂D. The conservation fluid variables $\mathbf{U}$ and flux vectors ($\mathbf{F}$, $\mathbf{G}$) are given by

$$\begin{aligned} \mathbf{U} &= \{\rho, \rho u, \rho v, E\}^T \\ \mathbf{F}(\mathbf{U}) &= \left\{\rho u, \rho u^2 + p, \rho uv, (E+p)u\right\}^T \\ \mathbf{G}(\mathbf{U}) &= \left\{\rho v, \rho uv, \rho v^2 + p, (E+p)v\right\}^T \end{aligned} \tag{3}$$

where ρ is the density, u and v are the velocity components in the x and y directions respectively, p is the pressure, and E is the total energy, which is the sum of the internal and kinetic energy:

$$E = \rho e + \frac{\rho}{2}(u^2 + v^2). \tag{4}$$

Considering a calorically perfect gas, the system of equations is completed via,

$$p = (\gamma - 1)\left\{E - \frac{\rho}{2}\left[u^2 + v^2\right]\right\} \tag{5}$$

Frequency Domain Equations

The next step in the HB development is to consider strictly periodic unsteady flows of a fundamental frequency ω. The flow variables and flux vectors are then approximated using truncated Fourier series in time with spatially varying coefficients. For instance,

$$\iint_{D(t)} \mathbf{U}\mathrm{d}D \approx \sum_{n=-N}^{N} \bar{\mathbf{U}}_n e^{in\omega t} \tag{6}$$

and

$$\begin{aligned} \oint_{\partial D(t)} \left(\mathbf{F}(\mathbf{U}) - \mathbf{U}\frac{\partial f}{\partial x}\right)\mathrm{d}y \\ - \oint_{\partial D(t)} \left(\mathbf{G}(\mathbf{U}) - \mathbf{U}\frac{\partial g}{\partial y}\right)\mathrm{d}x \approx \sum_{n=-N}^{N} \bar{\mathbf{F}}_n e^{in\omega t} \end{aligned} \tag{7}$$

where N is the number of harmonics used in the Fourier expansion. The time derivative of the vector of conservation fluid variables is then

$$\frac{\partial}{\partial t}\iint_{D(t)} \mathbf{U}\,\mathrm{d}D \approx in\omega \sum_{n=-N}^{N} \bar{\mathbf{U}}_n e^{in\omega t} \tag{8}$$

Substituting (7) and (8) into the Euler equations (2), multiplying by $e^{-im\omega t}$, and integrating over one period yields a system of equations for the Fourier coefficients

$$\mathbf{A}\bar{\mathbf{U}} + \bar{\mathbf{F}} = \mathbf{0} \tag{9}$$

where the diagonal matrix $\mathbf{A}$ and vectors $\bar{\mathbf{U}}$ and $\bar{\mathbf{F}}$ are

$$\begin{aligned} diag\mathbf{A} &= \{-iN, -i(N-1), \ldots, iN\} \\ \bar{\mathbf{U}} &= \left\{\bar{\mathbf{U}}_{-N}, \bar{\mathbf{U}}_{-(N-1)}, \ldots, \bar{\mathbf{U}}_N\right\}^T \\ \bar{\mathbf{F}} &= \left\{\bar{\mathbf{F}}_{-N}, \bar{\mathbf{F}}_{-(N-1)}, \ldots, \bar{\mathbf{F}}_N\right\}^T. \end{aligned} \tag{10}$$

As demostrated by Hall[5] via a Fourier transform matrix, $\mathbf{E}$, one can relate the Fourier coefficient variables, $\bar{\mathbf{U}}$, to the solution variables at $(2N+1)$ discrete time

levels within a period of motion. Hall also showed how one can express the Fourier Flux Coefficients in terms of time domain Flux term variables. This enables the HB methodology to be easily implemented within an existing steady CFD flow solver method. If one defines

$$\hat{\mathbf{U}} \equiv \left\{ \begin{array}{c} \hat{U}(t_0) \\ \cdots \\ \hat{U}(t_N) \end{array} \right\} \equiv \left\{ \begin{array}{c} \iint\limits_{D(t_0)} \mathbf{U}(t_0) \mathrm{d}D \\ \cdots \\ \iint\limits_{D(t_N)} \mathbf{U}(t_N) \mathrm{d}D \end{array} \right\} \tag{11}$$

and

$$\hat{\mathbf{F}} \equiv \left\{ \begin{array}{c} \hat{F}(t_0) \\ \cdots \\ \hat{F}(t_N) \end{array} \right\} \equiv$$
$$\left\{ \begin{array}{c} \oint\limits_{\partial D(t_0)} \left[\left(\mathbf{F}(\mathbf{U}) - \mathbf{U}\frac{\partial f}{\partial x} \right) \Big|_{t_0} \mathrm{d}y - \left(\mathbf{G}(\mathbf{U}) - \mathbf{U}\frac{\partial g}{\partial y} \right) \Big|_{t_0} \mathrm{d}x \right] \\ \cdots \\ \oint\limits_{\partial D(t_N)} \left[\left(\mathbf{F}(\mathbf{U}) - \mathbf{U}\frac{\partial f}{\partial x} \right) \Big|_{t_N} \mathrm{d}y - \left(\mathbf{G}(\mathbf{U}) - \mathbf{U}\frac{\partial g}{\partial y} \right) \Big|_{t_N} \mathrm{d}x \right] \end{array} \right\} \tag{12}$$

where

$$t_n = \frac{2\pi n}{(2N+1)\omega}, \quad n = 0, 1, ..., 2N \tag{13}$$

then

$$\bar{\mathbf{U}} = \mathbf{E}\hat{\mathbf{U}}, \quad \bar{\mathbf{F}} = \mathbf{E}\hat{\mathbf{F}} \tag{14}$$

Substituting Eqs. (14) into the equation for the Fourier coefficients, Eq. (9), yields

$$\mathbf{AE}\hat{\mathbf{U}} + \mathbf{E}\hat{\mathbf{F}} = \mathbf{0} \tag{15}$$

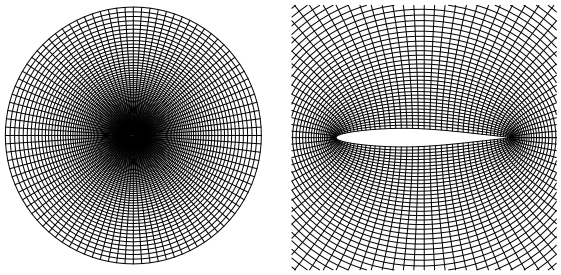
which after premultiplying by $\mathbf{E}^{-1}$ results in

$$\mathbf{D}\hat{\mathbf{U}} + \hat{\mathbf{F}} = \mathbf{0} \tag{16}$$

where $\mathbf{D} \equiv \mathbf{E}^{-1}\mathbf{AE}$ is the spectral source term operator that represents and approximates the temporal partial derivative, $\frac{\partial}{\partial t}$. As noted in Refs. [5,6], to solve the harmonic balance equations, one can add a pseudo time derivative term $\frac{\partial \hat{\mathbf{U}}}{\partial \tau}$ to Eq. (16):

$$\frac{\partial \hat{\mathbf{U}}}{\partial \tau} + \mathbf{D}\hat{\mathbf{U}} + \hat{\mathbf{F}} = \mathbf{0} \tag{17}$$

where τ is a fictitious time. Since only a "steady-state", temporally periodic solution is desired, local time stepping and multiple-grid acceleration techiniques can be used to increase the speed of computational convergence. The above described procedures allow one to model nonlinear unsteady aerodynamic of a prescribed frequency very efficiently. Again, see Refs. [5,6] for additional details.



a) Overall view **b)** Close-up

Fig. 2 Depiction of the 97×97, CFD Grid with Outer Boundary Radius of Ten Chord Lengths around a NACA 64A010A Airfoil.

Computational Grid

Fig. 2a shows the "O"-type computational grid used; a close-up in Fig. 2b shows the grid cells that surround a symmetric NACA 64A010A airfoil. The mesh consists of $N_r \times N_\Theta$ radial and circumferential nodes. Flutter and LCO results presented in this paper are computed using either 65×65 or 97×97 grids. However, grids of 25×25, 33×33 49×49 and 129×129 were also considered for a grid convergence study. The outer boundary radius or domain radius, R, was chosen to be either 5 or 10 chord lengths however the domains of $R/c = 15$ and $R/c = 20$ chord lengths were also considered. The dependence of the calculated aerodynamic forces and flutter boundaries on the grid and domain size was investigated to ensure convergence of the numerical results.

Aerodynamic Lift and Moment

For a simple harmonic motion of the airfoil,

$$\frac{h}{b} = \bar{h}e^{i\omega t}, \quad \alpha = \bar{\alpha}e^{i\omega t}. \tag{18}$$

the aerodynamic forces will usually be periodic in time, and for aeroelastic analysis the first harmonic will be dominant. Thus to a good approximation,

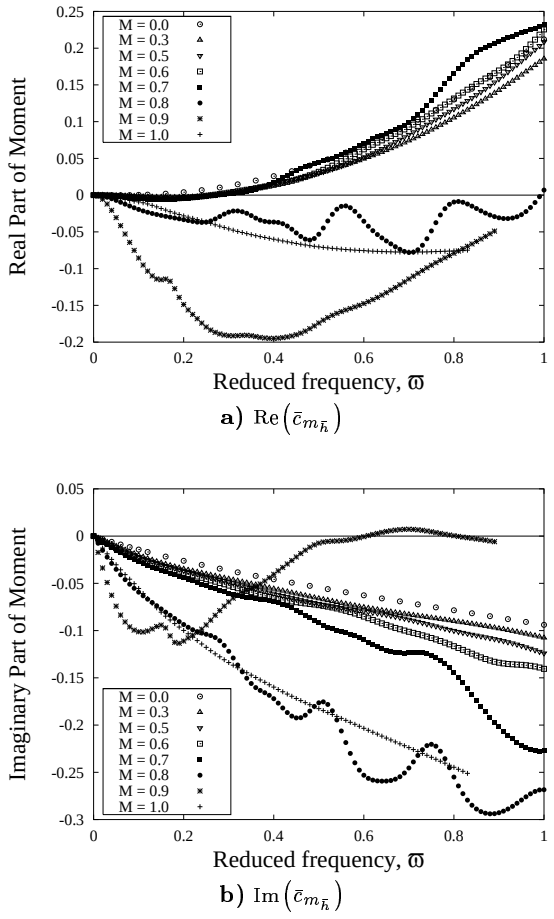
$$\begin{array}{rcl} L & = & \frac{\rho U_\infty^2 c}{2} \bar{c}_l \, e^{i\omega t} \\ M_{\text{e.a.}} & = & \frac{\rho U_\infty^2 c^2}{2} \bar{c}_m \, e^{i\omega t}. \end{array} \tag{19}$$

In general, the nondimensional lift and moment coefficients are expressed as

$$\begin{array}{l} \bar{c}_l = \bar{c}_l(\bar{\omega}, \text{geometry}, M, \bar{h}, \bar{\alpha}) \\ \bar{c}_m = \bar{c}_m(\bar{\omega}, \text{geometry}, M, \bar{h}, \bar{\alpha}), \end{array} \tag{20}$$

where $\bar{c}_l$ and $\bar{c}_m$ are nonlinear functions of $\bar{h}$ and $\bar{\alpha}$. For the LCO part of this study, in the calculation of aerodynamic lift and moment mostly two or three harmonics were used in the aerodynamic solution to *determine the effect of higher harmonics on the fundamental harmonic* in Eq. (19) for L and $M_{\text{e.a.}}$.

In dynamically or time linearized aerodynamics (used in flutter analyses per se), the nondimensional



a) $\mathrm{Re}\left(\bar{c}_{m_{\bar{h}}}\right)$ **b)** $\mathrm{Im}\left(\bar{c}_{m_{\bar{h}}}\right)$

Fig. 3 Real and Imaginary Parts of the Unsteady Aerodynamic Moment due to Plunge Motion vs Reduced Frequency.

lift and moment coefficients are linearly proportional to $\bar{h}$ and $\bar{\alpha}$:

$$\begin{aligned}\bar{c}_l &= \bar{c}_{l_{\bar{h}}}(\bar{\omega}, \text{geom}, M)\bar{h} + \bar{c}_{l_{\bar{\alpha}}}(\bar{\omega}, \text{geom}, M)\bar{\alpha} \\ \bar{c}_m &= \bar{c}_{m_{\bar{h}}}(\bar{\omega}, \text{geom}, M)\bar{h} + \bar{c}_{m_{\bar{\alpha}}}(\bar{\omega}, \text{geom}, M)\bar{\alpha}\end{aligned} \tag{21}$$

Some representative results for linear aerodynamic behavior computed by running the HB aerodynamic solver with very small amplitude motion ($\bar{h} = 0.001$) are presented in Fig. 3, which shows the real and imaginary parts of the aerodynamic moment due to the plunge motion, $\bar{c}_{m_{\bar{h}}}$, versus reduced frequency. For subsonic Mach numbers $0 < M < 0.7$, $\bar{c}_{m_{\bar{h}}}$ is relatively constant (on the reduced frequency interval $[0, 1]$). For $M = 0.8$ a "wavy" pattern is very prominent in $\bar{c}_{m_{\bar{h}}}$. This "wavy" pattern reflects the slow upstream acoustic waves typical of high subsonic/transonic flow.[8] The "wavy" pattern persists in the aerodynamic moment up to and including $M = 0.9$ (the Mach numbers of 0.82, 0.84, 0.86, and 0.88 were also studied, but the results are not shown in Fig. 3). At higher transonic numbers, from $M = 0.92$ and to $M = 1.0$, the shock location is at the trailing edge of the chord, the flow over the airfoil chord is supersonic, and the "wavy" pattern is no longer present. (Not plotted here are curves for Mach numbers of 0.92, 0.94, 0.96, and 0.98 as they correspond very closely to $M = 1.0$ results shown in Fig. 3.) The oscillatory behavior of $\bar{c}_{m_{\bar{h}}}$ is attributed to the tendency of the fluid dynamic eigenspectrum to become less damped for transonic Mach numbers. See Ref. [1].

Flutter

Rewriting the aeroelastic equations in terms of nondimensional variables yields

$$\left[-\bar{\omega}^2\begin{pmatrix}1 & x_\alpha \\ x_\alpha & r_\alpha^2\end{pmatrix} + \frac{4}{V^2}\begin{pmatrix}\frac{\omega_h^2}{\omega_\alpha^2} & 0 \\ 0 & r_\alpha^2\end{pmatrix} + \frac{4}{\pi\mu}\begin{pmatrix}\bar{c}_{l_{\bar{h}}} & \bar{c}_{l_{\bar{\alpha}}} \\ -2\bar{c}_{m_{\bar{h}}} & -2\bar{c}_{m_{\bar{\alpha}}}\end{pmatrix}\right]\begin{Bmatrix}\bar{h} \\ \bar{\alpha}\end{Bmatrix} = \mathbf{0}. \tag{22}$$

In functional form, to emphasize the parameters involved it can be written as

$$\left[\mathbf{F}(a, x_\alpha, r_\alpha, \text{geom}, M, \bar{\omega}, \mu, \frac{\omega_h}{\omega_\alpha}, V)\right]\left\{\begin{array}{c}\bar{h} \\ \bar{\alpha}\end{array}\right\} = \mathbf{0}. \tag{23}$$

Even though the airfoil model considered has only two DOF, there are numerous structural and aerodynamic parameters that will affect the behavior of $\mathbf{F}$.

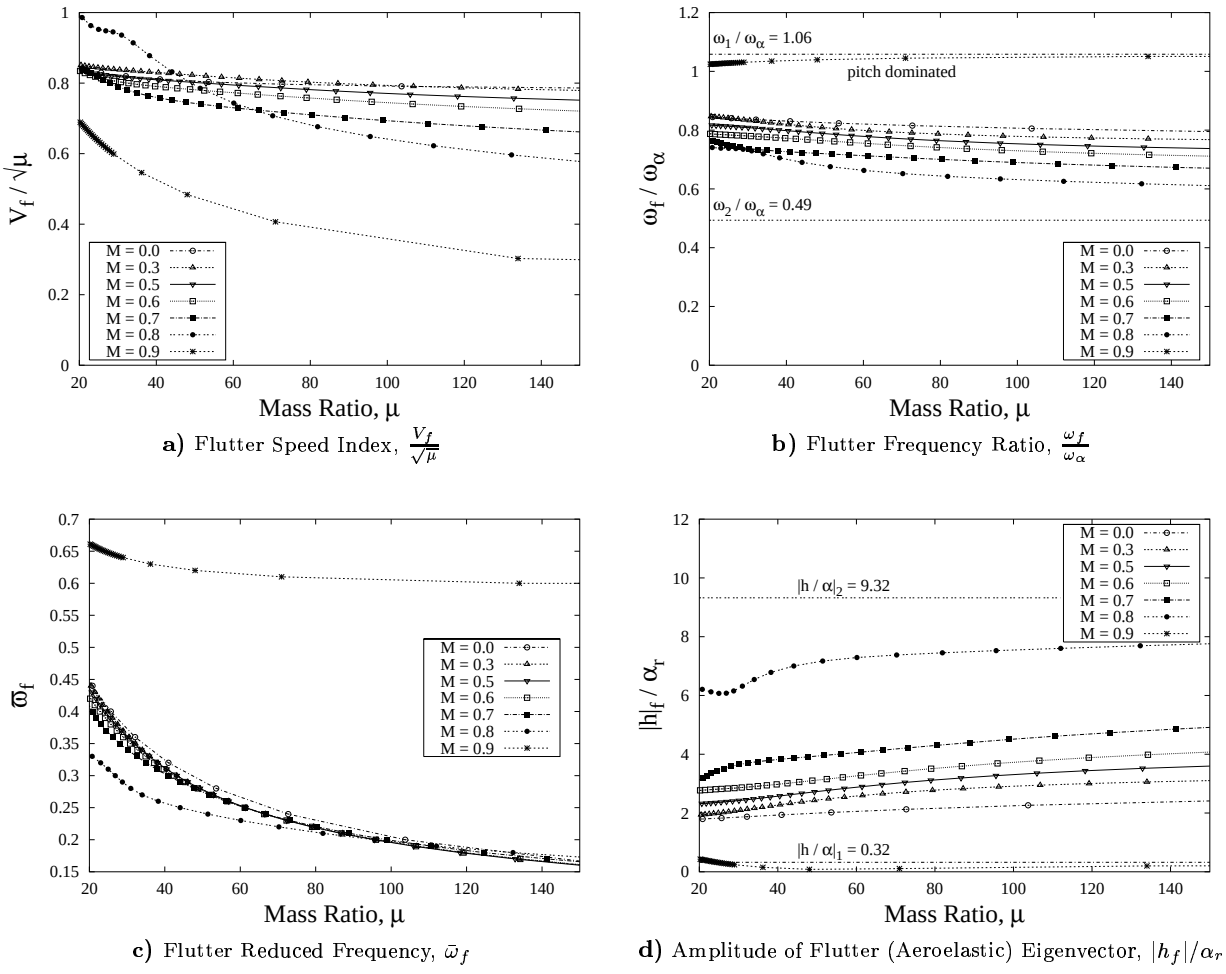
The elastic axis was placed at one-fifth of the chord ($a = -0.6$) and the airfoil radius of gyration and static unbalance were taken to be $r_\alpha^2 = 0.75$ and $x_\alpha = 0.25$ to conform to the studies of Ref. [6]. Aerodynamic coefficients, $\bar{c}_{l_{\bar{h}}}$, $\bar{c}_{l_{\bar{\alpha}}}$, etc., were computed for various Mach numbers for reduced frequencies between zero and one. Thus, this leaves only the mass ratio, μ, ratio of uncoupled frequencies, $\frac{\omega_h}{\omega_\alpha}$, and reduced velocity, V, as parameters to be considered in Eq. (23). Further fixing the value of either μ or $\frac{\omega_h}{\omega_\alpha}$, normalizing the structural mode shape $\{\bar{h}, \bar{\alpha}\}$ so as to have a purely real valued pitch component (the bars on top of nondimensional amplitudes $\bar{h}, \bar{\alpha}$ are omitted in what follows)

$$\left[\mathbf{F}(\frac{\omega_h}{\omega_\alpha} \text{ or } \mu, V)\right]\left\{\begin{array}{c}h_r + ih_i \\ \alpha_r\end{array}\right\} = \mathbf{0} \tag{24}$$

and setting the real and imaginary parts of the complex determinant to zero

$$det(F) = (0, 0), \tag{25}$$

one obtains two real equations in $\frac{1}{\mu}$ or $\frac{\omega_h^2}{{\omega_\alpha}^2}$ and $\frac{1}{V^2}$ (see Eq. (22)). The imaginary part of Eq. (25) provides a linear relation between the two unknowns, $\frac{1}{\mu}$ or $\frac{\omega_h^2}{{\omega_\alpha}^2}$ and $\frac{1}{V^2}$, and the real part of equation Eq. (25) then becomes a quadratic equation for $\frac{1}{\mu}$ or $\frac{\omega_h^2}{{\omega_\alpha}^2}$ that can be solved in closed form. The eigenvector, e.g. $(\frac{h_r}{\alpha_r}, \frac{h_i}{\alpha_r})$, at the flutter condition can then be determined from one of the two complex equations of (24).



a) Flutter Speed Index, $\frac{V_f}{\sqrt{\mu}}$

b) Flutter Frequency Ratio, $\frac{\omega_f}{\omega_\alpha}$

c) Flutter Reduced Frequency, $\bar{\omega}_f$

d) Amplitude of Flutter (Aeroelastic) Eigenvector, $|h_f|/\alpha_r$

Fig. 4 Flutter Boundary Behavior as a Function of the Mass Ratio for $\frac{\omega_h}{\omega_\alpha} = 0.5$.

Mass Ratio Effect

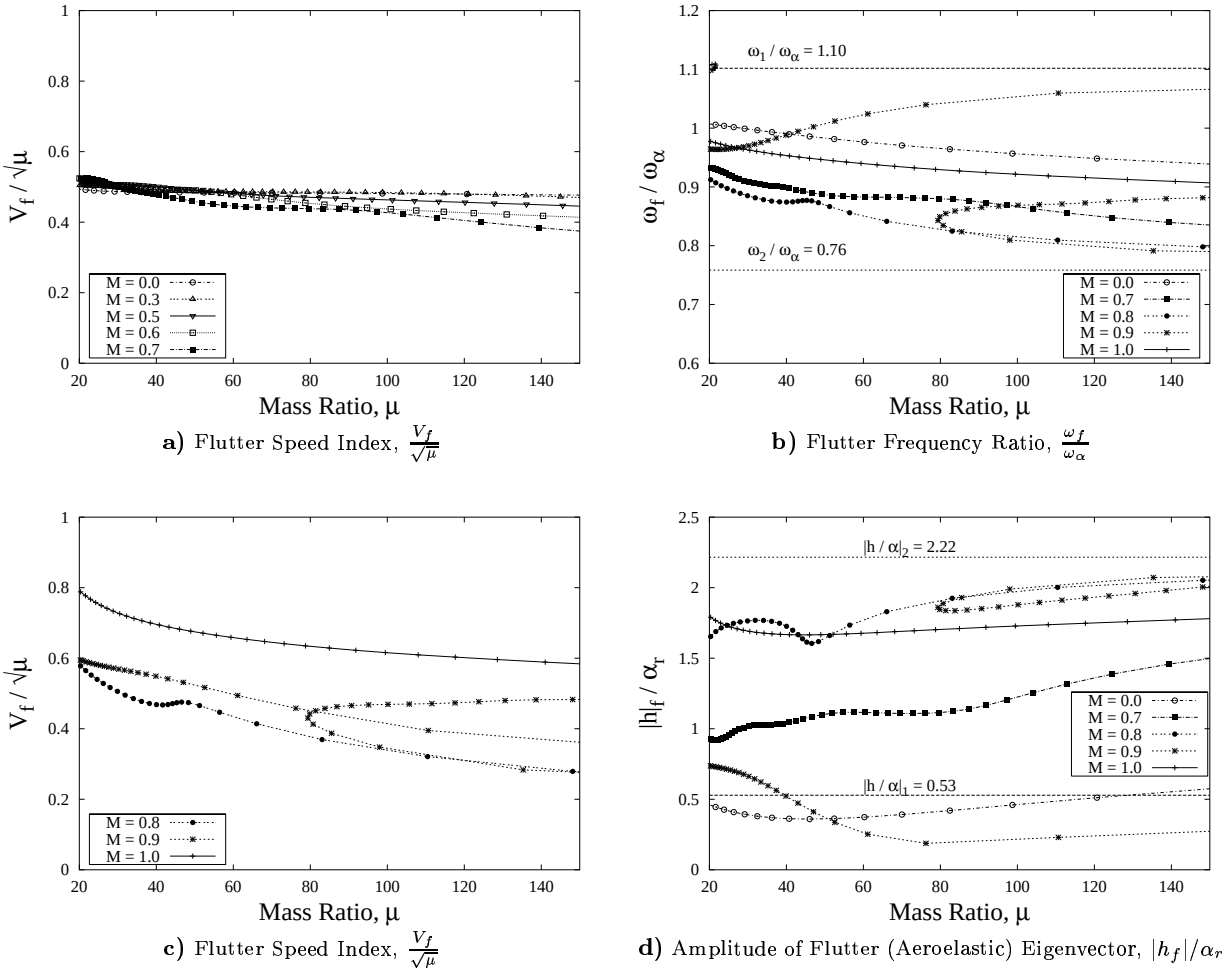
Dependence of the flutter boundaries (flutter speed index, flutter frequency and eigenvector) on the mass ratio, μ, was investigated first. Two cases for the ratio of uncoupled frequencies are presented here: $\frac{\omega_h}{\omega_\alpha} = 0.5$ (Fig. 4) and $\frac{\omega_h}{\omega_\alpha} = 0.8$ (Fig. 5).

$\frac{\omega_h}{\omega_\alpha} = 0.5$*:*

For $\frac{\omega_h}{\omega_\alpha} = 0.5$, the flutter speed index (Fig. 4a) is seen to be weakly dependent on μ for subsonic Mach numbers. For transonic Mach numbers of $M = 0.8$ and $M = 0.9$ the flutter speed index decreases with μ at a much higher rate. For $M = 1.0$ the flutter speed index becomes very large (well beyond the scale of the graph). Note that the flutter reduced frequency (Fig. 4c) for $M = 0.9$ is distinctly higher than for other Mach numbers. The flutter frequency (Fig. 4b) and flutter eigenvector (Fig. 4d) are also less sensitive to mass ratio at low subsonic Mach numbers with a more significant dependency occurring at higher subsonic and transonic Mach numbers. Coupled in vacuo natural frequencies, ω_1 and ω_2, were also computed for easy reference and as can be seen in Fig. 4b, the flutter motion is pitch dominated for $M = 0.9$. One may also see this from Fig. 4d, where coupled in vacuo eigenvectors, $(\frac{h}{\alpha})_1$ and $(\frac{h}{\alpha})_2$, are indicated. The flutter eigenvector for $M = 0.9$ is essentially the same as $(\frac{h}{\alpha})_1$, which corresponds to the pitch dominated mode.

$\frac{\omega_h}{\omega_\alpha} = 0.8$*:*

For $\frac{\omega_h}{\omega_\alpha} = 0.8$, the flutter speed index (Fig. 5a) is even more weakly dependent on μ for subsonic Mach numbers. In Fig. 5c, the flutter speed index behavior for transonic Mach numbers of $M = 0.8$, $M = 0.9$ and $M = 1.0$ is shown. As in the case of $\frac{\omega_h}{\omega_\alpha} = 0.8$, the flutter speed index is much more sensitive to μ than for the lower, subsonic Mach numbers. For this mass ratio however, when $M = 0.9$, multiple flutter velocities occur for mass ratios $\mu \geq 80$. To determine the stability of the regions created by such multiple branches of flutter boundaries a root-loci analysis for fixed values of $\frac{\omega_h}{\omega_\alpha}$ and μ can be employed.[9–11] However in this study only flutter boundaries were determined. The flutter frequency (Fig. 5b) and flutter eigenvector (Fig. 5d) are distributed over the full range between coupled natural frequencies and eigenvectors and none



a) Flutter Speed Index, $\frac{V_f}{\sqrt{\mu}}$

b) Flutter Frequency Ratio, $\frac{\omega_f}{\omega_\alpha}$

c) Flutter Speed Index, $\frac{V_f}{\sqrt{\mu}}$

d) Amplitude of Flutter (Aeroelastic) Eigenvector, $|h_f|/\alpha_r$

Fig. 5 Flutter Boundary Behavior as a Function of the Mass Ratio for $\frac{\omega_h}{\omega_\alpha} = 0.8$.

of Mach number cases appears to be primarily pitch or plunge dominated.

Ratio of Uncoupled Natural Frequencies Effect

Dependence of the flutter boundary on the ratio of uncoupled natural frequencies, $\frac{\omega_h}{\omega_\alpha}$, is next investigated, and two mass ratios are considered: $\mu = 25$ (Fig. 6 and 7) and $\mu = 100$ (Fig. 8).

$\mu = 25$:

For $\mu = 25$, the flutter speed index (Fig. 6a) has a minimum near $\frac{\omega_h}{\omega_\alpha} = 1$. For typical airfoils, $\frac{\omega_h}{\omega_\alpha}$ is in general between 0.2 and 0.8. In that range, the flutter speed index decreases steadily and is almost constant for Mach numbers less than 0.7. However, as can be seen and is well known, the flutter speed index becomes very sensitive to the Mach number for high subsonic or transonic conditions.

For a better understanding of how the flutter speed index results are obtained one can show that the flutter reduced velocity may be expressed as

$$V_f^2 = \frac{d_1 \frac{\omega_h^2}{\omega_\alpha^2} + d_2}{d_3}, \tag{26}$$

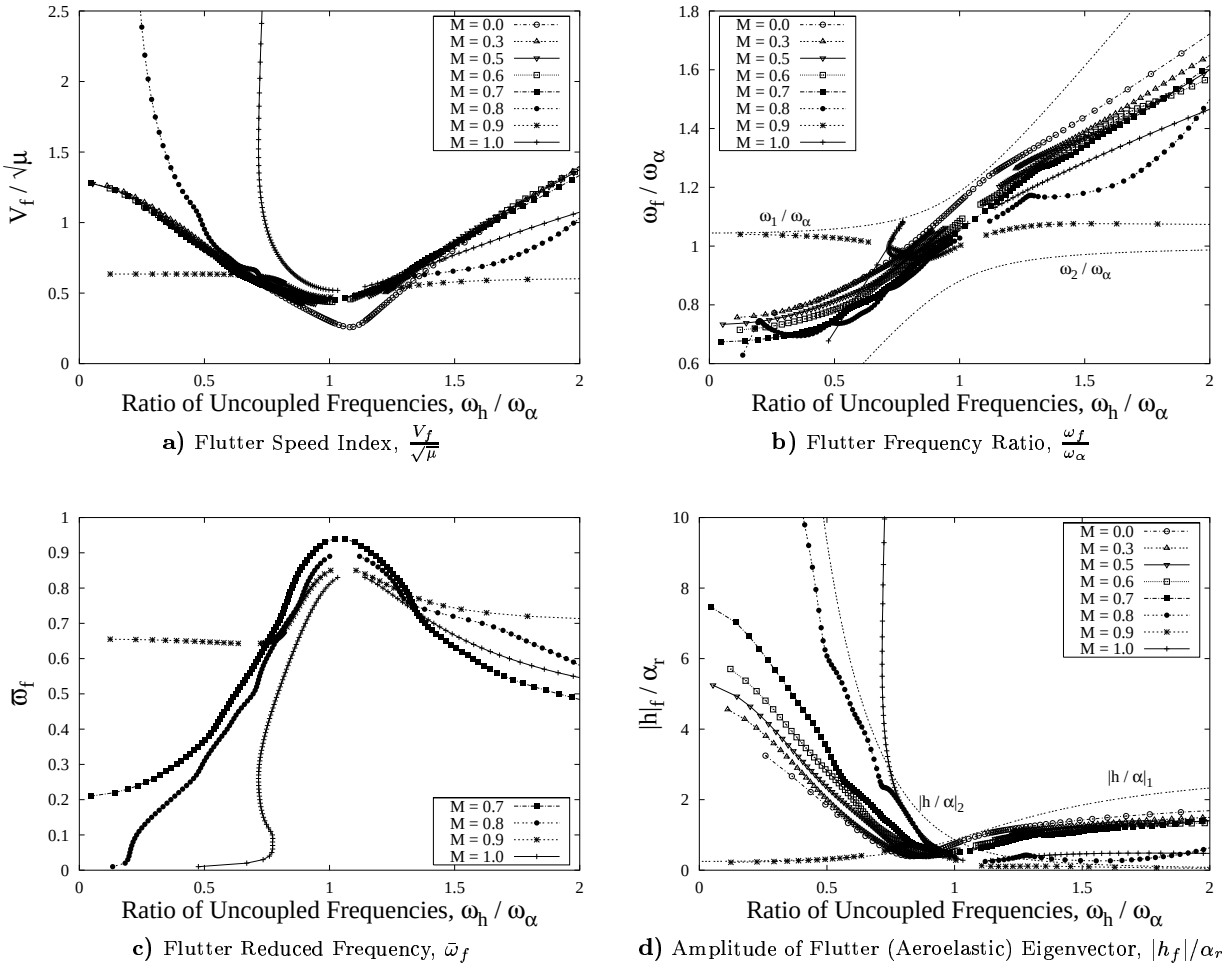
where d_1, d_2, and d_3 are coefficients dependent on the reduced frequency. Eq. (26) is obtained from the imaginary part of Eq. (25). Substitution of Eq. (26) into the real part of Eq. (25), leads to a quadratic equation in $\frac{\omega_h^2}{\omega_\alpha^2}$,

$$a_2 \left(\frac{\omega_h^2}{\omega_\alpha^2}\right)^2 + a_1 \frac{\omega_h^2}{\omega_\alpha^2} + a_0 = 0 \tag{27}$$

with the roots given by

$$\left(\frac{\omega_h^2}{\omega_\alpha^2}\right)_{1,2} = \frac{-a_1 \pm \sqrt{a_1^2 - 4a_2 a_0}}{2a_2}, \tag{28}$$

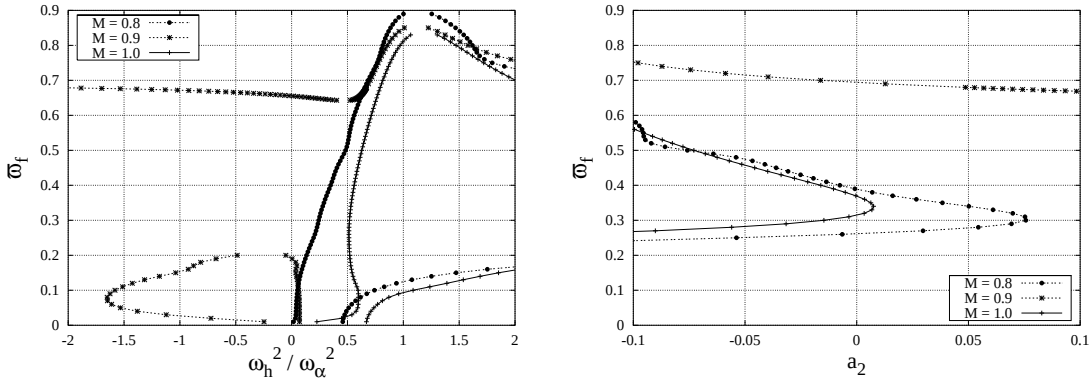
where the coefficients a_i may also change the sign for different $\bar{\omega}$. Obviously, only positive real values of $\frac{\omega_h^2}{\omega_\alpha^2}$ in Eq. (28) have physical meaning. Moreover, only such values of $\frac{\omega_h^2}{\omega_\alpha^2}$ are kept for which V_f^2 in Eq. (26) is



a) Flutter Speed Index, $\frac{V_f}{\sqrt{\mu}}$ **b)** Flutter Frequency Ratio, $\frac{\omega_f}{\omega_\alpha}$

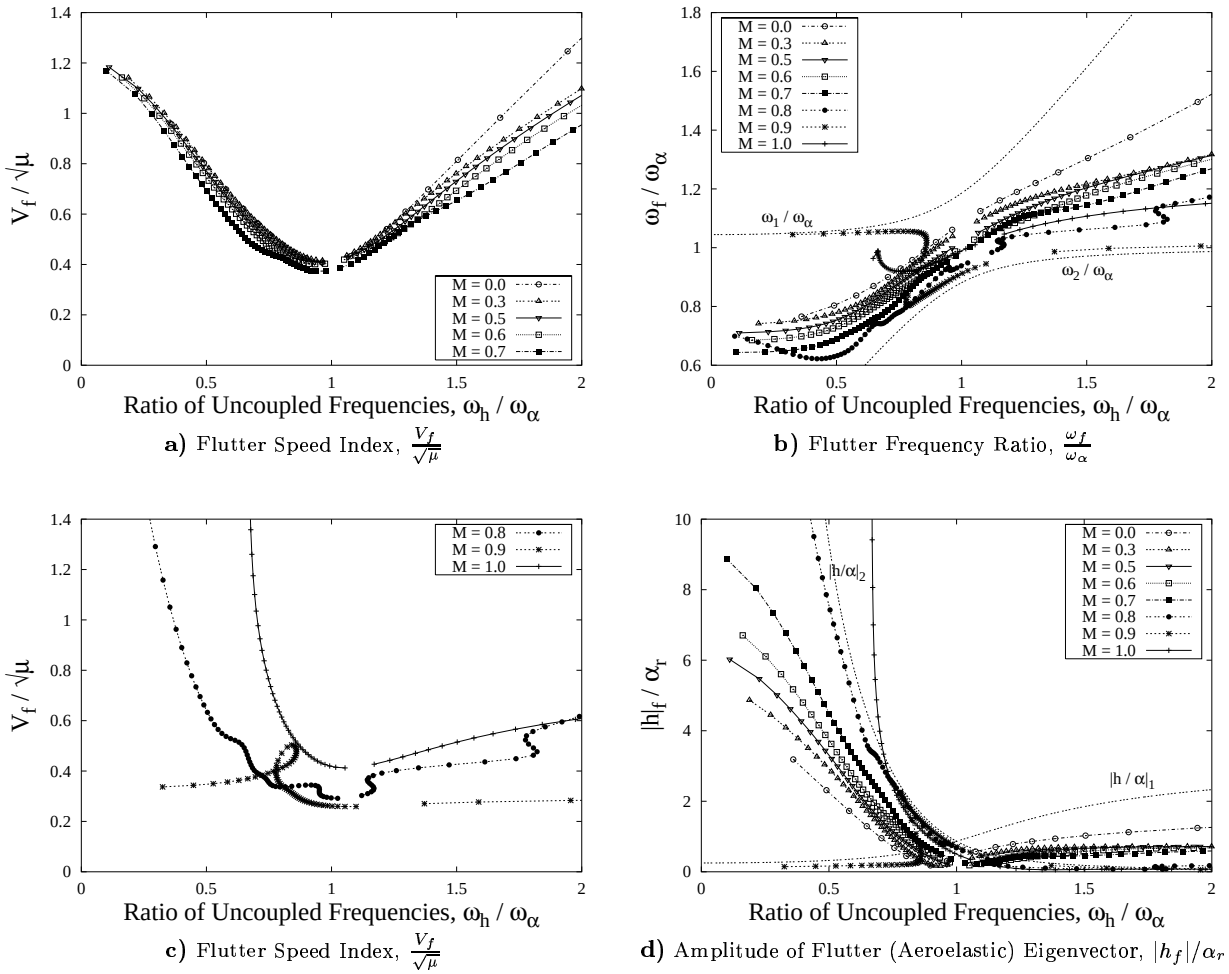
c) Flutter Reduced Frequency, $\bar{\omega}_f$ **d)** Amplitude of Flutter (Aeroelastic) Eigenvector, $|h_f|/\alpha_r$

Fig. 6 Flutter Boundary Behavior as a Function of the Ratio of Uncoupled Frequencies for $\mu = 25$.



a) **b)**

Fig. 7 Flutter Reduced Frequency, $\bar{\omega}_f$, versus a) $\frac{\omega_h^2}{\omega_\alpha^2}$ and b) the coefficient a_2 of $a_2 (\frac{\omega_h^2}{\omega_\alpha^2})^2 + a_1 \frac{\omega_h^2}{\omega_\alpha^2} + a_0 = 0$ for $\mu = 25$.



a) Flutter Speed Index, $\frac{V_f}{\sqrt{\mu}}$ **b)** Flutter Frequency Ratio, $\frac{\omega_f}{\omega_\alpha}$

c) Flutter Speed Index, $\frac{V_f}{\sqrt{\mu}}$ **d)** Amplitude of Flutter (Aeroelastic) Eigenvector, $|h_f|/\alpha_r$

Fig. 8 Flutter Boundary Behavior as a Function of the Ratio of Uncoupled Frequencies for $\mu = 100$.

positive. Real roots of Eq. (28) are shown in Fig. 7a. When the coefficient a_2 passes through zero, branches of $\frac{\omega_h^2}{\omega_\alpha^2}$ assymptotically go to $\pm\infty$. The dependence of a_2 with respect to $\bar{\omega}_f$ is shown in Fig. 7b. For example, in Fig. 6c for $M = 0.9$, a_2 goes through zero at $\bar{\omega}_f \simeq 0.7$, which is when the branches of $\frac{\omega_h^2}{\omega_\alpha^2}$ go to $\pm\infty$ in Fig. 7a.

Note that the difference between the two coupled natural frequencies (ω_1 and ω_2 as indicated in Fig. 6b) is the smallest near $\frac{\omega_h}{\omega_\alpha} = 1$. This is likely responsible for the minimum in the flutter speed index in the vicinity of that point: the closer the coupled natural frequencies are initially, the more readily, when increasing velocity, "coalescence" or "merging frequency" flutter can occur (see p. 112 of Ref. [8]).

$\mu = 100$:

For $\mu = 100$, the flutter speed index (Fig. 8a and c) again has the tendency to have a minimum near $\frac{\omega_h}{\omega_\alpha} = 1$, and especially so for subsonic Mach numbers (Fig. 8a). As in the case for $\mu = 25$, the flutter speed index is nearly constant with M for subsonic conditions. At higher transonic Mach numbers (Fig. 8c), the flutter speed index becomes very sensitive to Mach number. This behavior is assosiated with the "transonic flutter dip". Also note that the flutter speed index can be a multivalued function of $\frac{\omega_h}{\omega_\alpha}$ for $M = 0.8$ and $M = 0.9$. Note that (Fig. 8b) the flutter frequency is very close to the coupled natural pitch frequency, $\frac{\omega_1}{\omega_\alpha}$, for $M = 0.9$ (pitch dominated motion). This is indicative of "single-degree-of-freedom" flutter (see p. 113 of Ref. [8]).

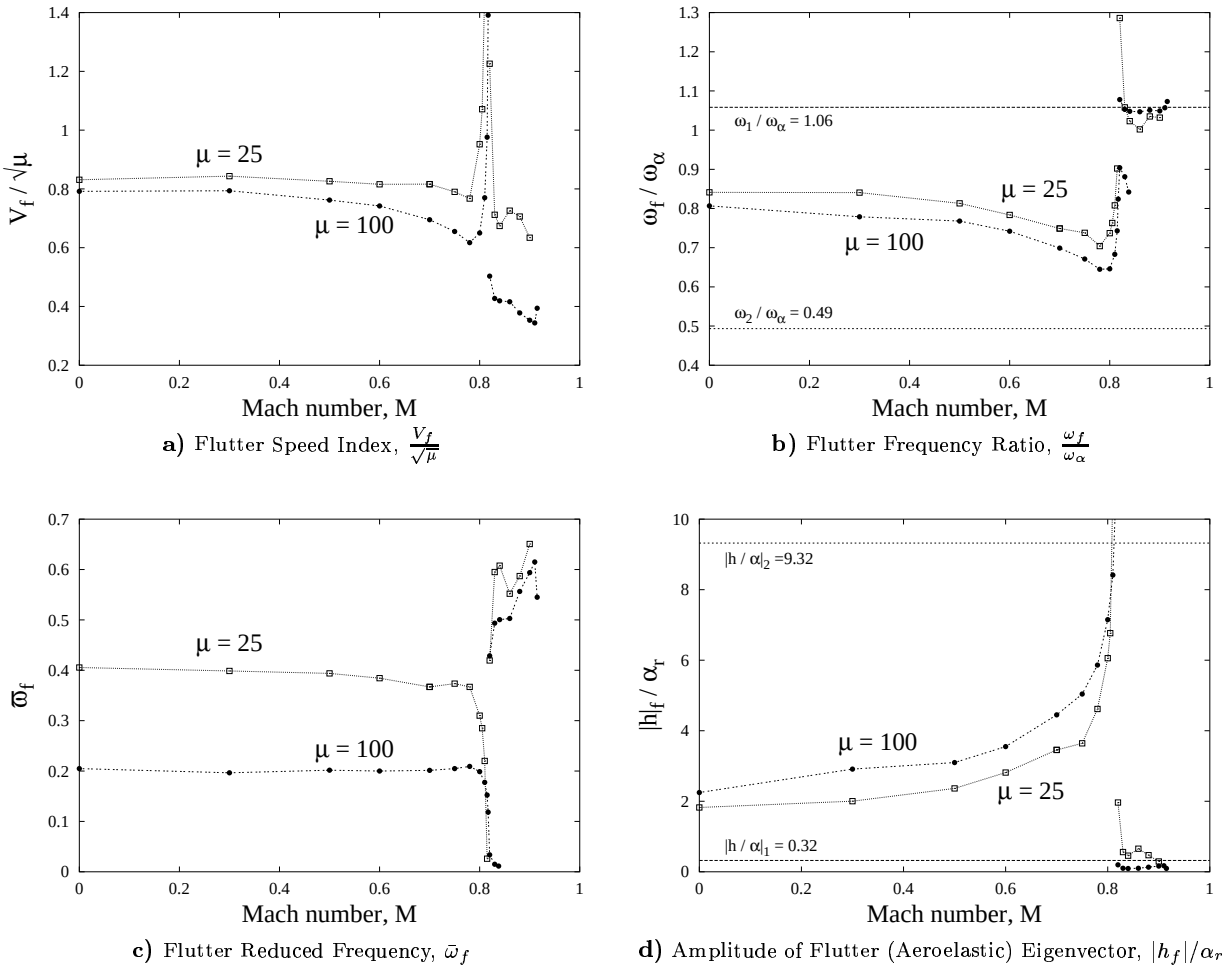
Mach Number Effect

The dependence of the flutter boundary on Mach number (see Figs. 9 and 10) for the cases of $\mu = 25$, 100 and $\frac{\omega_h}{\omega_\alpha} = 0.5$, 0.8 is presented next.

$\frac{\omega_h}{\omega_\alpha} = 0.5$:

Fig. 9 shows the flutter boundary behavior versus Mach number for $\frac{\omega_h}{\omega_\alpha} = 0.5$. Blank squares represent the case of $\mu = 25$ and filled circles the case of $\mu = 100$. Note a rather weak dependence of the flutter boundaries on the mass ratio.

For $\mu = 100$, as the Mach number is increased



a) Flutter Speed Index, $\frac{V_f}{\sqrt{\mu}}$

b) Flutter Frequency Ratio, $\frac{\omega_f}{\omega_\alpha}$

c) Flutter Reduced Frequency, $\bar{\omega}_f$

d) Amplitude of Flutter (Aeroelastic) Eigenvector, $|h_f|/\alpha_r$

Fig. 9 Flutter Boundary Behavior as a Function of Mach number for $\frac{\omega_h}{\omega_\alpha} = 0.5$ and $\mu = 25,\ 100$.

past $M = 0.8$, a sharp drop in the flutter reduced frequency occurs. That is accompanied by a sharp increase in the flutter speed index. As the Mach number is increased further, another branch of flutter points appears. These low flutter speed index values correspond to the "transonic flutter dip".

For an explanation to the abrupt change in flutter mode at $M = 0.82$ and $M = 0.915$ one may ask whether the steady flow shock position is changing and where the shock location is relative to the elastic axis and mode nodes.

The steady flow pressure contours for $M = 0.81$ and $M = 0.82$ did not show any significant changes in the pressure distribution. However the situation is different for Mach numbers $M = 0.91$ and $M = 0.92$. Here the location of the shock moves to the trailing edge as the Mach number is increased to $M = 0.92$. For $\frac{\omega_h}{\omega_\alpha} = 0.8$, the flutter behavior also changes in the viscinity of these Mach numbers. Note that in the Mach number range $0.82 < M < 0.92$ (Fig.11) the position of the shock on the airfoil moves appreciably. To develop an improved understanding of the behavior of the flutter boundaries one could perform a root locus analysis which would enable one to view the migration of the aeroelastic eigenvalues as a function of the nondimensional airspeed for each Mach number. Representative root locus analyses for transonic Mach numbers of similar airfoil models can be found in Refs. [9–11]. Root loci however have not been part of the present study.

The reduced flutter frequency behavior is shown in Fig. 9c. For subsonic and low transonic Mach numbers ($0 \leq M \leq 0.8$), $\bar{\omega}_f$ is much higher for the lower mass ratio ($\bar{\omega}_f \approx 0.4$) than for the higher mass ratio ($\bar{\omega}_f \approx 0.2$). At high transonic Mach numbers ($0.82 \leq M < 0.92$), the flutter reduced frequency is about the same $0.45 \leq \bar{\omega}_f \leq 0.65$ for either mass ratio.

The flutter (aeroelastic) eigenvector is shown in Fig. 9d and provides a possible explanation for the low flutter speed index values for the transonic dip Mach number region. In this range, the flutter mode becomes pitch dominated for both values of mass ratio. The flutter mode is markedly different from that for $M < 0.82$, for example.

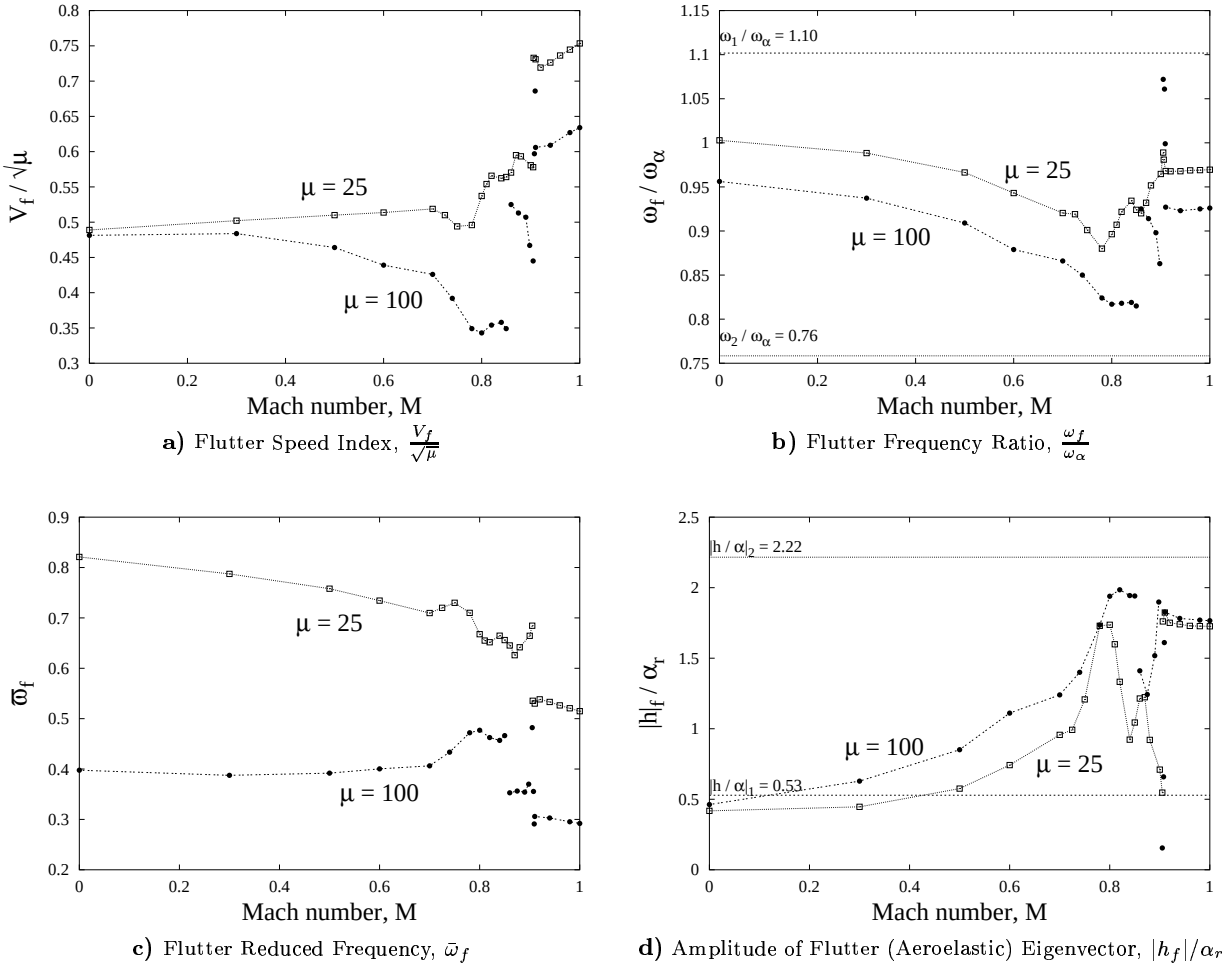


Fig. 10 Flutter Boundary Behavior as a Function of Mach number for $\frac{\omega_h}{\omega_\alpha} = 0.8$ and $\mu = 25,\ 100$.

$\frac{\omega_h}{\omega_\alpha} = 0.8$*:*

The flutter speed index for $\frac{\omega_h}{\omega_\alpha} = 0.8$ is shown in Fig. 10a. For this rather high value of pitch to plunge frequency ratio, there is no transonic dip in the flutter speed index at the lower mass ratio. Moreover the system appears to be more stable at the high transonic Mach numbers. For the higher mass ratio as the Mach number increases, the flutter velocity index reaches its minimum in the range $0.78 < M < 0.85$. The flutter mode then appears to change twice for even higher Mach numbers. Contrary to the results for $\frac{\omega_h}{\omega_\alpha} = 0.5$, here the flutter frequency shown in Fig. 10b is not very close to either the pitch or plunge coupled natural frequency. So the flutter mode (Fig. 10d) is not a single degree flutter motion, but rather a combined pitch-plunge motion.

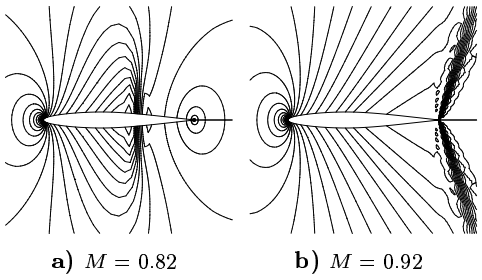


Fig. 11 Steady Flow Pressure Contours.

Modeling LCO Behavior

The nonlinear aeroelastic equations of motion in the frequency domain for an arbitrary amplitude of motion can be written[6] as

$$\left[-\bar{\omega}^2\begin{pmatrix}1 & x_\alpha\\ x_\alpha & r_\alpha^2\end{pmatrix}+\frac{4}{V^2}\begin{pmatrix}\frac{\omega_h^2}{\omega_\alpha^2} & 0\\ 0 & r_\alpha^2\end{pmatrix}\right]\begin{Bmatrix}\bar{h}\\ \bar{\alpha}\end{Bmatrix}=\frac{4}{\pi\mu}\begin{Bmatrix}-\bar{c}_l(\bar{h},\bar{\alpha},\bar{\omega})\\ 2\,\bar{c}_m(\bar{h},\bar{\alpha},\bar{\omega})\end{Bmatrix}. \tag{29}$$

Specifying a real valued amplitude of pitch DOF as the independent variable, $\bar{\alpha}$, the real and imaginary parts of Eq. (29) then constitute a system of four nonlinear

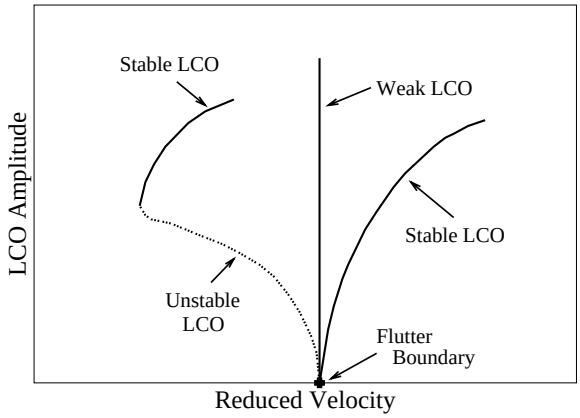


Fig. 12 Generic Display of the LCO Behavior.

real equations for the vector of unknowns

$$\mathbf{L} = \left\{ \bar{\omega}, V, (\frac{\bar{h}}{\bar{\alpha}})_{Re}, (\frac{\bar{h}}{\bar{\alpha}})_{Im} \right\}^T . \tag{30}$$

Eq. (29) can then be written in the vector form $\mathbf{R}(\mathbf{L}) = \mathbf{0}$, or

$$\begin{bmatrix} -\bar{\omega}^2(\frac{\bar{h}_r}{\bar{\alpha}})_{Re} + \frac{4}{V^2}\frac{\omega_h^2}{\omega_\alpha^2}(\frac{\bar{h}}{\bar{\alpha}})_{Re} - \bar{\omega}^2 x_\alpha + \frac{4}{\pi\mu\bar{\alpha}}(\bar{c}_c)_{Re} \\ -\bar{\omega}^2(\frac{\bar{h}}{\bar{\alpha}})_{Im} + \frac{4}{V^2}\frac{\omega_h^2}{\omega_\alpha^2}(\frac{\bar{h}}{\bar{\alpha}})_{Im} + \frac{4}{\pi\mu\bar{\alpha}}(\bar{c}_l)_{Im} \\ -\bar{\omega}^2 x_\alpha(\frac{\bar{h}}{\bar{\alpha}})_{Re} \quad -\bar{\omega}^2 r_\alpha^2 + \frac{4}{V^2} r_\alpha^2 - \frac{8}{\pi\mu\bar{\alpha}}(\bar{c}_m)_{Re} \\ -\bar{\omega}^2 x_\alpha(\frac{\bar{h}}{\bar{\alpha}})_{Im} - \frac{8}{\pi\mu\bar{\alpha}}(\bar{c}_m)_{Im} \end{bmatrix} = \mathbf{0}. \tag{31}$$

A Newton-Raphson method was used to solve the system, Eq. (31).

$$\mathbf{L}^{n+1} = \mathbf{L}^n - \left[\frac{\partial \mathbf{R}(\mathbf{L}^n)}{\partial \mathbf{L}} \right]^{-1} \mathbf{R}(\mathbf{L}^n) \tag{32}$$

where the Jacobian matrix $\frac{\partial \mathbf{R}(\mathbf{L}_n)}{\partial \mathbf{L}}$ can be approximated using forward differences.[6] For example,

$$\begin{aligned} \left.\frac{\partial \mathbf{R}(\mathbf{L}^n)}{\partial \mathbf{L}}\right|_{1,1} &\equiv \frac{\partial \mathbf{R}_1(\mathbf{L}^n)}{\partial \bar{\omega}} \approx \frac{\mathbf{R_1}(\mathbf{L}^n, \bar{\omega}^n + \epsilon) - \mathbf{R_1}(\mathbf{L}^n, \bar{\omega}^n)}{\epsilon} \\ \left.\frac{\partial \mathbf{R}(\mathbf{L}^n)}{\partial \mathbf{L}}\right|_{1,2} &\equiv \frac{\partial \mathbf{R}_1(\mathbf{L}^n)}{\partial V} \approx \frac{\mathbf{R_1}(\mathbf{L}^n, V^n + \epsilon) - \mathbf{R_1}(\mathbf{L}^n, V^n)}{\epsilon} \end{aligned} \tag{33}$$

for a small ϵ. Thus during each iteration the HB flow solver is run four times: one time to determine $\mathbf{R}$ for $\mathbf{L}^n$, and three more times to determine $\mathbf{R}$ for $\mathbf{L}^n$ with sensitivities in $\bar{\omega}$, $(\frac{\bar{h}}{\bar{\alpha}})_{Re}$ and $(\frac{\bar{h}}{\bar{\alpha}})_{Im}$. Once the values of Newton-Raphson's method sequence are sufficiently close to the root, the convergence is so rapid that only a few more values are needed.

Using the flutter reduced velocity, frequency and eigenvector obtained from a linearized aerodynamic model as starting solution one can then calculate a LCO solution for a specified small amplitude of pitch $\bar{\alpha}$ in just a few iterations. The process can then be repeated for a larger LCO amplitude. Smaller amplitude increments must be used for successful convergence if the nonlinear effect is very strong.

One of the main objectives of this study was to determine the effects of aerodynamic nonlinearity of the flow on LCO behavior of the airfoil. The dependence of LCO behavior on the Mach number presents the greatest challenge especially for transonic Mach numbers.

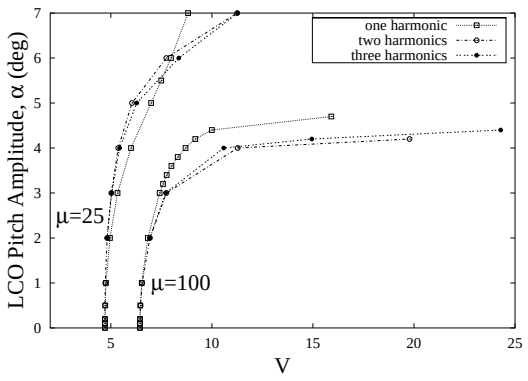
In presenting the results, a distinction will be made with respect to stable LCO, unstable LCO, and weak (stable or unstable) LCO. To explain these terms, consider Fig. 12, which displays three generic curves for the amplitude of LCO vs reduced velocity. If the aerodynamic model is dynamically linear, then in this limiting case the LCO amplitude vs reduced velocity curve would simply be a vertical line indicating that the steady limit cycle amplitude is infinite when the flutter boundary and the corresponding reduced velocity is reached and exceeded. Hence a vertical or near vertical graph of LCO amplitude vs reduced velocity is said to be a "weak" LCO. Note that a weak LCO in fact corresponds to a large LCO amplitude. Conversely if the LCO amplitude is smaller and the graph of LCO amplitude vs reduced velocity bends more to the right or left it is said to be a "strong" LCO. Note that weak or strong refers to the strength of the (aerodynamic) nonlinearity and the LCO amplitude itself is correspondingly large or small.

Now the other distinction to be emphasized is that between stable and unstable LCO. If the LCO amplitude increases with increasing reduced velocity beyond the flutter boundary, then the LCO is said to be stable. Physically this means that LCO does not occur below the flutter boundary and that the LCO that occurs above the flutter boundary is stable with respect to small infinitesimal disturbances. Conversely if the LCO amplitude increases as the reduced velocity decreases from the flutter boundary, then the LCO is said to be unstable since LCO may now occur for reduced velocities below the flutter boundary. Moreover, a stability analysis for the LCO per se with respect to infinitesimal disturbances will usually show that any perturbation to this unstable LCO will lead to a transient motion that at large time will reach a new LCO that is stable (typically at a larger amplitude). Again see Fig. 12 for a generic display of this latter behavior. In the present work a perturbation stability analysis of the LCO per se has not been carried out and the identification of stable or unstable LCO is made based solely on the dependence of LCO amplitude on reduced velocity as shown in Fig. 12.

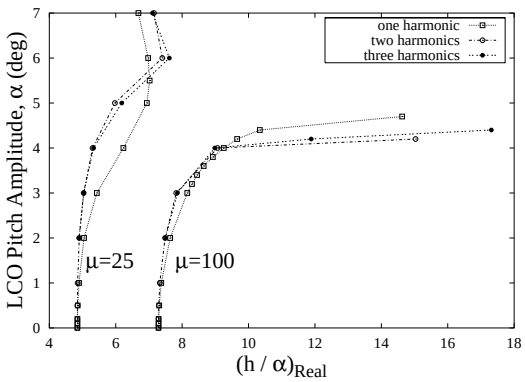
The discussion of the LCO results begins with the dependence of LCO on mass and uncoupled natural frequencies ratios at Mach number $M = 0.8$.

Ratio of Uncoupled Natural Frequencies and Mass Ratio Effect

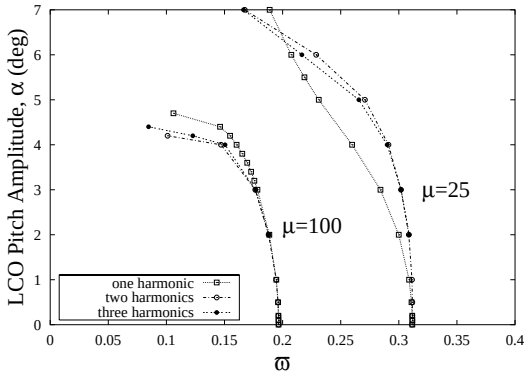
LCO results are presented for four ratios of uncoupled frequencies ($\frac{\omega_h}{\omega_\alpha} = 0.5$, 0.8, 1.5, and 1.8). In each



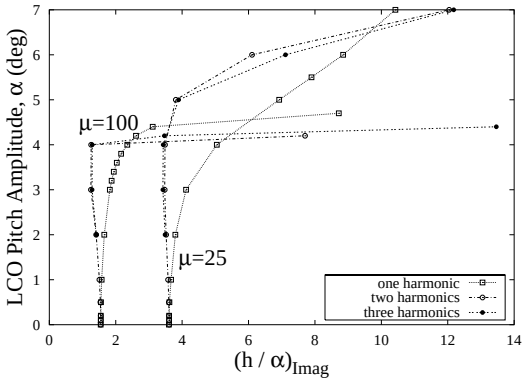
a) LCO Reduced Velocity



b) Real Part of LCO (Aeroelastic) Eigenvector

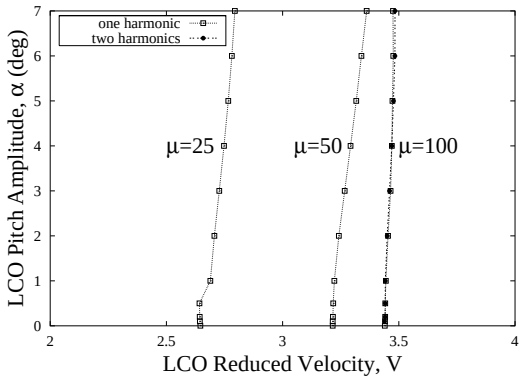


c) LCO Reduced Frequency

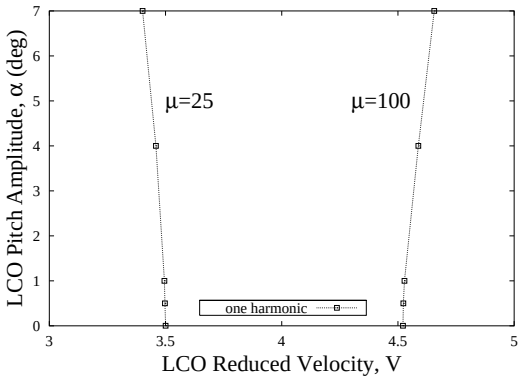


d) Imaginary Part of LCO (Aeroelastic) Eigenvector

Fig. 13 LCO Behavior for $\frac{\omega_h}{\omega_\alpha} = 0.5$ and $M = 0.8$.

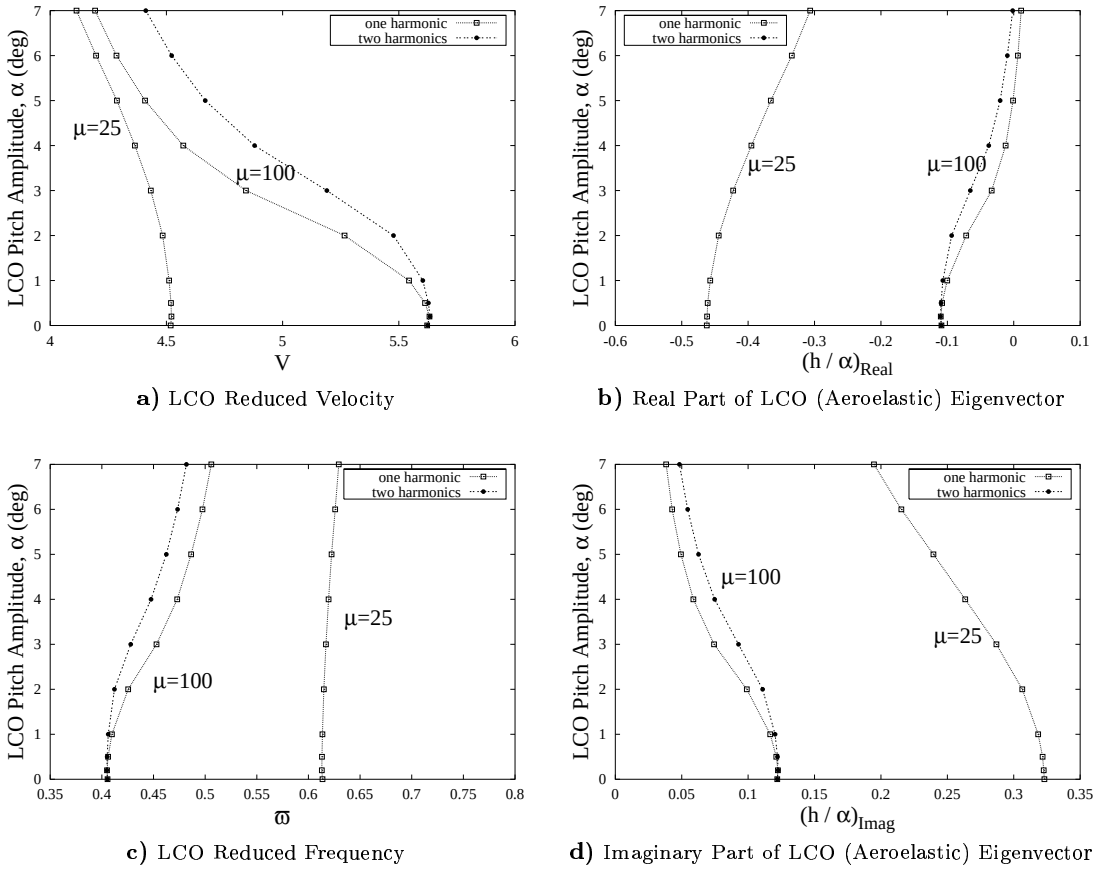


a) $\frac{\omega_h}{\omega_\alpha} = 0.8$



b) $\frac{\omega_h}{\omega_\alpha} = 1.5$

Fig. 14 LCO Behavior for $\frac{\omega_h}{\omega_\alpha} = 0.8$, 1.5 and $M = 0.8$.



a) LCO Reduced Velocity

b) Real Part of LCO (Aeroelastic) Eigenvector

c) LCO Reduced Frequency

d) Imaginary Part of LCO (Aeroelastic) Eigenvector

Fig. 15 LCO Behavior for $\frac{\omega_h}{\omega_\alpha} = 1.8$ and $M = 0.8$.

case, two (μ = 25, 100) or three (μ = 25, 50 and 100) mass ratios were considered for a Mach number of $M = 0.8$. Also considered was harmonic convergence of the computed unsteady aerodynamic forces via the HB solver. A distinct transition from stable LCO to weak LCO and then to unstable LCO as the ratio of uncoupled frequencies is increased will be noted. In all cases, a higher mass ratio is found to correspond to a higher reduced velocity for a particular LCO amplitude and a lower reduced frequency. A 65×65 CFD grid with $\frac{R}{c} = 10$ chord lengths was used.

$\frac{\omega_h}{\omega_\alpha} = 0.5$*:*

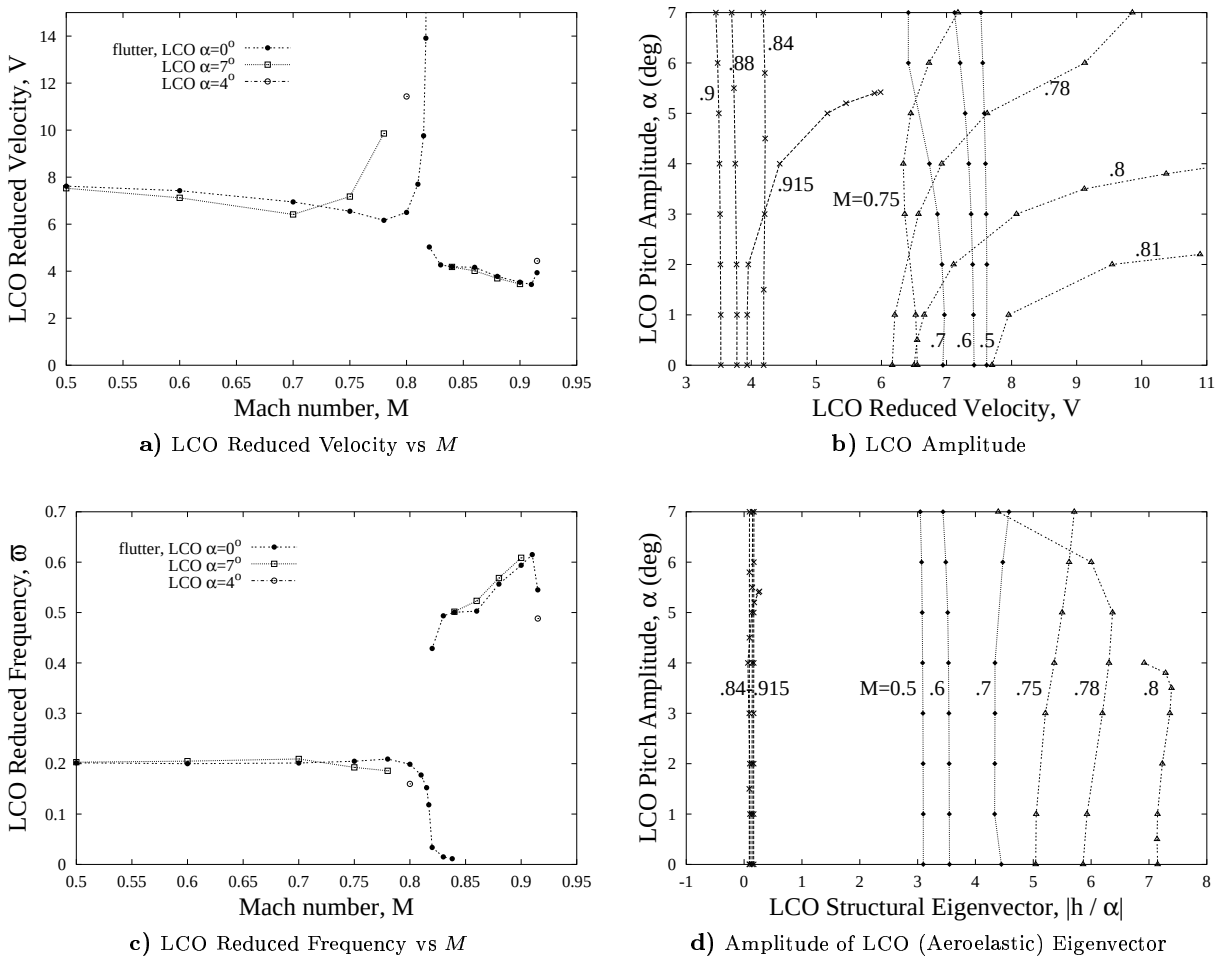
Fig. 13a shows the case of a stable LCO in terms of LCO amplitude vs reduced velocity. Employing just two harmonics in the HB solver provides a good level of harmonic convergence for moderate LCO amplitudes. The motion is plunge dominant at low amplitudes and becomes even more so ($|\frac{\bar{h}}{\bar{\alpha}}|$ increases) for large LCO amplitudes.

$\frac{\omega_h}{\omega_\alpha} = 0.8$ *and* $\frac{\omega_h}{\omega_\alpha} = 1.5$*:*

Fig. 14 shows LCO results for frequency ratios of $\frac{\omega_h}{\omega_\alpha}$ = 0.8 and 1.5. In both cases the nonlinearity is rather weak – the LCO curves depend weakly on the amplitude, i. e. they are close to being straight vertical lines. The present HB solver has some numerical stability problems for large reduced frequencies, which in turn means that using multiple harmonics for such high frequencies may not work. For the case of $\frac{\omega_h}{\omega_\alpha} = 0.8$ and $\frac{\omega_h}{\omega_\alpha} = 1.5$, the LCO reduced frequency is indeed quite high ($0.5 \leq \bar{\omega} \leq 0.7$), and thus only one harmonic could be used in most instances in the HB solver. However, the authors believe, and the case of $\mu = 100$ in Fig. 14a (where two harmonics were used) supports this, that when the nonlinearity is weak, one harmonic used in the computations of aerodynamic lift and moment can give a reliable LCO result. There is a difference in the amplitude of the eigenvector between these two frequencies (not shown here): for $\frac{\omega_h}{\omega_\alpha} = 0.8$, $|\frac{\bar{h}}{\bar{\alpha}}|$ is of the order of one – the motion is complex pitch-plunge motion, while for $\frac{\omega_h}{\omega_\alpha} = 1.5$, it is predominantly pitch motion.

$\frac{\omega_h}{\omega_\alpha} = 1.8$*:*

For this rather high frequency ratio (see Fig. 15), the LCO is unstable (for this Mach number at least), especially for the higher mass ratio. The aeroelastic LCO eigenmodes are very much pitch dominanted as might be expected. Thus it can be concluded that an increase in the ratio of uncoupled frequency ratio (from $\frac{\omega_h}{\omega_\alpha} = 0.5$ to $\frac{\omega_h}{\omega_\alpha} = 1.8$) is accompanied



a) LCO Reduced Velocity vs M

b) LCO Amplitude

c) LCO Reduced Frequency vs M

d) Amplitude of LCO (Aeroelastic) Eigenvector

Fig. 16 LCO Behavior with respect to Mach Number for $\frac{\omega_h}{\omega_\alpha} = 0.5$ and $\mu = 100$.

by changes in stablity of LCO (from stable through weak to unstable) and in aeroelastic eigenmode (from plunge dominance through pitch-plunge complex motion to pitch dominance). One can also conlcude that a higher mass ratio gives rise to a larger nonlinear effect for LCO at this Mach number. As far as LCO velocity, reduced frequency and eigenvalues are concerned, the parametric trends presented in the flutter portion of this study continue to be observed for the LCO solution per se as well. For example, for the same μ the reduced velocity (for a specified LCO amplitude) first decreases and then increases as $\frac{\omega_h}{\omega_\alpha}$ is increased with the minimum point being in the vicinity of $\frac{\omega_h}{\omega_\alpha} = 1$.

Mach Number Effect

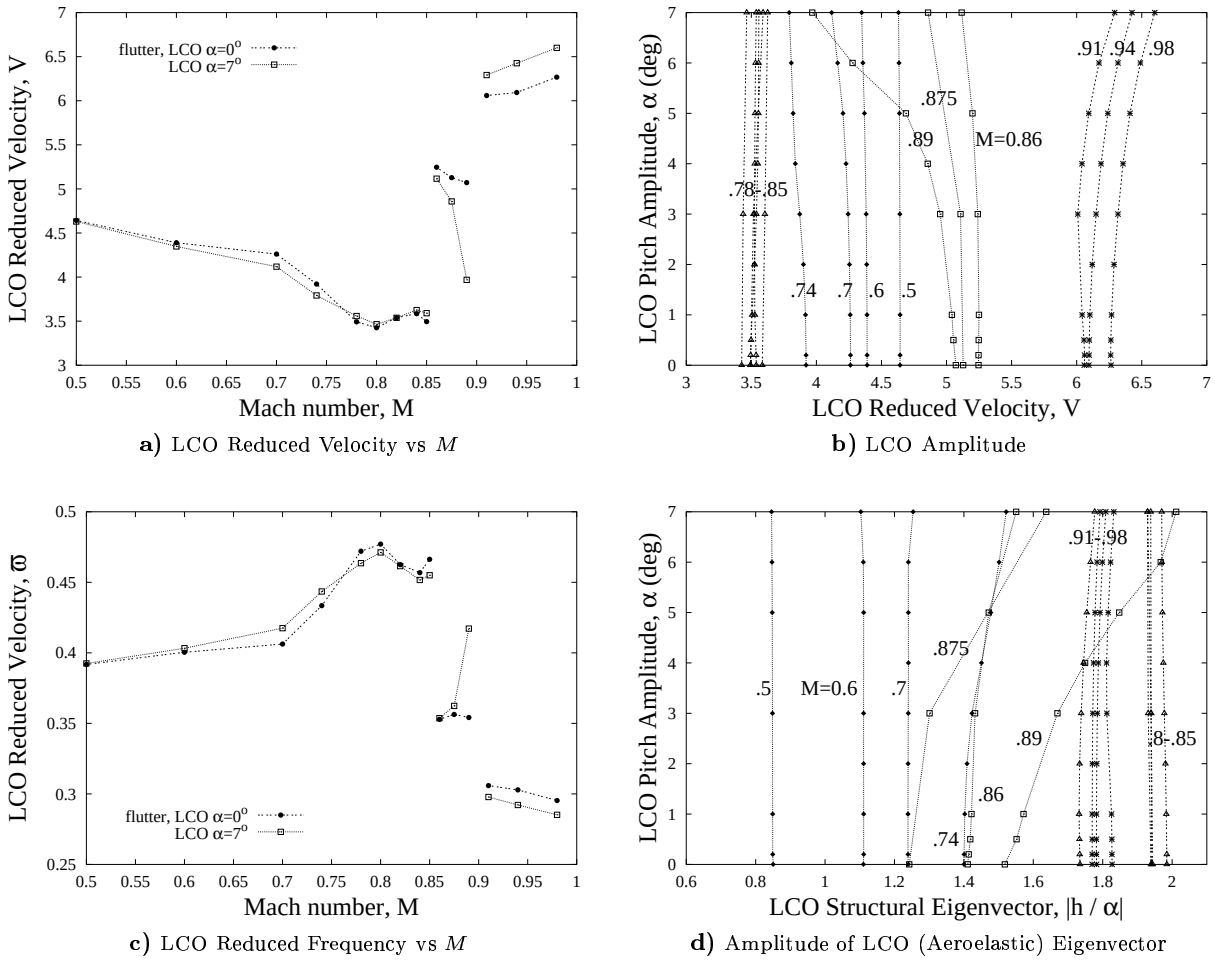
LCO behavior as a function of Mach number is next studied for two ratios of uncoupled frequencies, $\frac{\omega_h}{\omega_\alpha} = 0.5$ and 0.8 for a single mass ratio, $\mu = 100$ (Figs. 16 and 17). Reduced velocity and reduced frequency vs Mach number trends are shown for various LCO amplitudes including the flutter boundary corresponding to an LCO amplitude of zero. These results allow one to observe the stability of the LCO over a wide range of Mach numbers. Note in particular, that when the LCO reduced velocity is above the flutter reduced velocity, the LCO is stable, and when it is below the LCO is unstable. LCO behavior for two frequency ratios are now presented.

$\frac{\omega_h}{\omega_\alpha} = 0.5$*:*

The LCO velocity vs Mach number results in Fig. 16a and b show that in the range of $0.78 < M < 0.81$ the aerodynamic nonlinearity effects are strong. In the interval Mach number range $0.5 < M < 0.7$ the LCO is unstable but only weakly so with the motion being plunge dominated (see LCO eigenvectors in Fig. 16d). For $0.75 < M < 0.81$, the LCO becomes highly stable with predominantly plunge type motion. Finally, for $0.84 < M < 0.9$ the nonlinearity is very weak and the motion is mostly pitch dominated.

$\frac{\omega_h}{\omega_\alpha} = 0.8$*:*

For this ratio of uncoupled frequencies (Fig. 17), there is also only a small range of Mach numbers where aerodynamic nonlinearity is relatively significant. However, it is interesting to note the numerous



a) LCO Reduced Velocity vs M

b) LCO Amplitude

c) LCO Reduced Frequency vs M

d) Amplitude of LCO (Aeroelastic) Eigenvector

Fig. 17 LCO Behavior with respect to Mach Number for $\frac{\omega_h}{\omega_\alpha} = 0.8$ and $\mu = 100$.

changes in the stability of the LCO over the range of Mach numbers considered. LCO are unstable for $0.5 < M < 0.74$, stable for $0.78 < M < 0.85$, unstable again for $0.86 < M < 0.89$, and finally LCO are stable again for $0.91 < M < 0.98$. It is not very surprising because of the weak LCO that the LCO eigenvectors are little different from the flutter eigenvectors. The results in Fig. 17d show that the motion is complex pitch-plunge motion just as was concluded from the flutter results for the corresponding case, Fig. 10d.

Conclusions

Using a state of the art Euler CFD based aerodynamic code, an investigation is presented of how structural and aerodynamic parameters including freestream (transonic) Mach number affect flutter and LCO characteristics of a "typical" two degree-of-freedom airfoil configuration.

The following conclusions have been drawn:

I When examining the dependence of the flutter and LCO on the mass ratio, μ, it was found that the flutter and LCO solution variables (i.e. reduced velocity, frequency and structural eigenmode) vary little with μ, especially for subsonic Mach numbers.

II As expected, a study of the effect of the ratio of uncoupled natural frequencies, $\frac{\omega_h}{\omega_\alpha}$, determined that flutter and LCO reduced velocities have a minimum near $\frac{\omega_h}{\omega_\alpha} \approx 1$. Multiple values of the flutter velocity can occur at high transonic Mach numbers for some $\frac{\omega_h}{\omega_\alpha}$.

III Most significantly perhaps, it was demonstrated that flutter and LCO solutions are very sensitive to Mach number especially in the transonic range.

(a) Depending on the frequency and the mass ratios, there may or may not be sudden and significant changes (e.g. the "transonic dip") in the flutter reduced velocity and the type of flutter and LCO motion as the Mach number is varied.

(b) The strength of the aerodynamic nonlinearity changes with M. Moreover it was found that this nonlinearity is most prominent in

a limited range of transonic Mach numbers and rather weak outside this Mach number range. The stability of LCO was also found to abruptly change with respect to Mach number.

An extensive parameter analyses of an airfoil aeroelastic configuration has been achieved using a highly efficient HB computational technique. Correlation with experiment remains an open challenge.

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